

## 2.2 Evaluating Determinants by Row Reduction

*In this section we show how to evaluate a determinant by reducing the associated matrix to row echelon form. This method generally requires less computation than cofactor expansion and is the method of choice for large matrices.*

### A Basic Theorem

#### Theorem 2.2.1

Let  $A$  be a square matrix. If  $A$  has a row of zeros or a column of zeros, then  $\det(A) = 0$ .

**Proof.** Expand along the zero row or column; every term is  $0 \cdot C_{ij} = 0$ . ■

#### Theorem 2.2.2

Let  $A$  be a square matrix. Then  $\det(A) = \det(A^T)$ .

**Proof.** Transposing  $A$  swaps its rows and columns; thus a cofactor expansion of  $A$  along any row equals the cofactor expansion of  $A^T$  along the corresponding column. ■

**Remark.** Because  $\det(A) = \det(A^T)$ , every theorem about row operations on determinants has an analogous column version.

### Elementary Row Operations

The following theorem describes how each type of elementary row (or column) operation affects  $\det(A)$ .

#### Theorem 2.2.3

Let  $A$  be an  $n \times n$  matrix, and let  $B$  result from a single elementary row (or column) operation on  $A$ .

- (a) **Scale:** If a single row (or column) of  $A$  is multiplied by a scalar  $k$ , then  $\det(B) = k \det(A)$ .
- (b) **Interchange:** If two rows (or columns) of  $A$  are interchanged, then  $\det(B) = -\det(A)$ .
- (c) **Row addition:** If a multiple of one row (or column) of  $A$  is added to another, then  $\det(B) = \det(A)$ .

The following table illustrates Theorem 2.2.3 for the  $3 \times 3$  case.

Table 1

| Relationship                                                                                                                                                                                                                                                                        | Operation                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| $\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = k \det(A)$                        | In $B$ , the first row of $A$ was multiplied by $k$ .        |
| $\begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = -\det(A)$                            | In $B$ , the first and second rows of $A$ were interchanged. |
| $\begin{vmatrix} a_{11} + ka_{21} & a_{12} + ka_{22} & a_{13} + ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ $\det(B) = \det(A)$ | In $B$ , a multiple of row 2 of $A$ was added to row 1.      |

**Proof of (a).** Both determinants on the two sides of the first equation share the same cofactors  $C_{11}, C_{12}, C_{13}$  along the first row (cofactors depend only on the *other* rows). Expanding the left side along row 1:

$$\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ \dots & \dots & \dots \end{vmatrix} = ka_{11}C_{11} + ka_{12}C_{12} + ka_{13}C_{13} = k(a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}) = k \det(A). \quad \blacksquare$$

## Elementary Matrices

When  $A = I_n$  in Theorem 2.2.3,  $B$  is an elementary matrix  $E$ . The theorem then gives:

**Theorem 2.2.4**

Let  $E$  be an  $n \times n$  elementary matrix.

- (a) If  $E$  results from multiplying a row of  $I_n$  by  $k$ , then  $\det(E) = k$ .
- (b) If  $E$  results from interchanging two rows of  $I_n$ , then  $\det(E) = -1$ .
- (c) If  $E$  results from adding a multiple of one row of  $I_n$  to another, then  $\det(E) = 1$ .

**Remark.** The determinant of an elementary matrix is *never* zero.

**EXAMPLE 1 | Determinants of Elementary Matrices**

The following  $4 \times 4$  elementary matrices illustrate Theorem 2.2.4:

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 3, \quad \begin{vmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} = -1, \quad \begin{vmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1$$

(Row 2 of  $I_4$  multiplied by 3; first and last rows of  $I_4$  interchanged; 7 times last row added to first row.)

**Matrices with Proportional Rows or Columns**

If  $A$  has two proportional rows, adding a suitable multiple of one to the other produces a zero row. Since row addition does not change  $\det$  (Theorem 2.2.3(c)), and a zero-row matrix has  $\det = 0$  (Theorem 2.2.1), we get:

**Theorem 2.2.5**

If a square matrix  $A$  has two proportional rows or two proportional columns, then  $\det(A) = 0$ .

**EXAMPLE 2 | Proportional Rows or Columns**

Each of the following has two proportional rows or columns, so each has determinant zero:

$$\begin{bmatrix} -1 & 4 \\ -2 & 8 \end{bmatrix}, \quad \begin{bmatrix} 1 & -2 & 7 \\ -4 & 8 & 5 \\ 2 & -4 & 3 \end{bmatrix}, \quad \begin{bmatrix} 3 & -1 & 4 & -5 \\ 6 & -2 & 5 & 2 \\ 5 & 8 & 1 & 4 \\ -9 & 3 & -12 & 15 \end{bmatrix}$$

(In the first, row 2 is  $2 \times$  row 1; in the second, row 2 is  $-4 \times$  row 1 while cols 1 and 2 are proportional ( $-2 \times$ ); in the third, row 4 is  $-3 \times$  row 1.)

**Evaluating Determinants by Row Reduction**

The strategy: reduce  $A$  to upper triangular form by EROs (tracking the sign changes from interchanges and scalar factors), then multiply the diagonal entries.

**EXAMPLE 3 | Using Row Reduction to Evaluate a Determinant**

Evaluate  $\det(A)$  where  $A = \begin{bmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{bmatrix}$ .

**SOLUTION:**

$$\begin{aligned}
 \det(A) &= \begin{vmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{vmatrix} = - \begin{vmatrix} 3 & -6 & 9 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix} && \text{(rows 1 \& 2 interchanged)} \\
 &= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix} && \text{(factor 3 from row 1)} \\
 &= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{vmatrix} && \text{(-2 \cdot row 1 added to row 3)} \\
 &= -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{vmatrix} && \text{(-10 \cdot row 2 added to row 3)} \\
 &= (-3)(-55) \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{vmatrix} && \text{(factor -55 from row 3)} \\
 &= (-3)(-55)(1) = \boxed{165}
 \end{aligned}$$

**EXAMPLE 4 | Using Column Operations to Evaluate a Determinant**

Compute  $\det(A)$  where  $A = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 2 & 7 & 0 & 6 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -5 \end{bmatrix}$ .

**SOLUTION:** Add  $-3$  times column 1 to column 4 to put  $A$  in lower triangular form:

$$\det(A) = \det \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 7 & 0 & 0 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -26 \end{bmatrix} = (1)(7)(3)(-26) = \boxed{-546}$$

**EXAMPLE 5 | Row Operations and Cofactor Expansion**

Evaluate  $\det(A)$  where  $A = \begin{bmatrix} 3 & 5 & -2 & 6 \\ 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \\ 3 & 7 & 5 & 3 \end{bmatrix}$ .

**SOLUTION:** Add multiples of row 2 to the other rows to create zeros in column 1:

$$\det(A) = \begin{vmatrix} 0 & -1 & 1 & 3 \\ 1 & 2 & -1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 1 & 8 & 0 \end{vmatrix}$$

Cofactor expand along column 1 (only  $a_{21} = 1$  is nonzero):

$$= -1 \cdot (-1)^{2+1} \begin{vmatrix} -1 & 1 & 3 \\ 0 & 3 & 3 \\ 1 & 8 & 0 \end{vmatrix} = - \begin{vmatrix} -1 & 1 & 3 \\ 0 & 3 & 3 \\ 1 & 8 & 0 \end{vmatrix}$$

Add row 1 to row 3:

$$= - \begin{vmatrix} -1 & 1 & 3 \\ 0 & 3 & 3 \\ 0 & 9 & 3 \end{vmatrix}$$

Cofactor expand along column 1:

$$= -[(-1)(-1)^{1+1} \begin{vmatrix} 3 & 3 \\ 9 & 3 \end{vmatrix}] = -[(-1)(9 - 27)] = -(18) = \boxed{-18}$$

## Exercise Set 2.2

In Exercises 1-4, verify that  $\det(A) = \det(A^T)$ .

1.  $A = \begin{bmatrix} -2 & 3 \\ 1 & 4 \end{bmatrix}$

3.  $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \\ 5 & -3 & 6 \end{bmatrix}$

In Exercises 5-8, find the determinant of the given elementary matrix by inspection.

5.  $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

7.  $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

In Exercises 9-13, evaluate the determinant of the matrix by first reducing to row echelon form and then using some combination of row operations and cofactor expansion.

9.  $\begin{bmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{bmatrix}$

$$11. \begin{bmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

$$13. \begin{bmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

In Exercises 15–22, evaluate the determinant, given that  $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$ .

$$15. \begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix}$$

$$17. \begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix}$$

$$20. \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g+3a & h+3b & i+3c \end{vmatrix}$$

$$22. \begin{vmatrix} a & b & c \\ d & e & f \\ 2a & 2b & 2c \end{vmatrix}$$

23. Use row reduction to show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

24. Verify the formulas in parts (a) and (b) and then conjecture a general result.

$$(a) \det \begin{bmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = -a_{13}a_{22}a_{31}$$

$$(b) \det \begin{bmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = a_{14}a_{23}a_{32}a_{41}$$

In Exercises 25 and 27, confirm the identity without evaluating any of the determinants directly.

$$25. \begin{vmatrix} a_1 + b_1 & b_1 + c_1 & c_1 \\ a_2 + b_2 & b_2 + c_2 & c_2 \\ a_3 + b_3 & b_3 + c_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$27. \begin{vmatrix} a_1 + b_1 & a_1 - b_1 & c_1 \\ a_2 + b_2 & a_2 - b_2 & c_2 \\ a_3 + b_3 & a_3 - b_3 & c_3 \end{vmatrix} = -2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

In Exercises 29–30, show that  $\det(A) = 0$  without directly evaluating the determinant.

$$29. A = \begin{bmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & 3 \end{bmatrix}$$

33. Let  $A$  be an  $n \times n$  matrix, and let  $B$  be the matrix that results when the rows of  $A$  are written in reverse order. State a theorem that describes how  $\det(A)$  and  $\det(B)$  are related.

34. Find the determinant of the matrix

$$\begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}$$